

FTML practical session 12

solution

3 juin 2026

Solution of the Lasso regression
1 dimension, 1 sample
 $\theta_\lambda = \operatorname{argmin}_\theta \frac{1}{2}(y - \theta x)^2 + \lambda|\theta|$
 $x=1$
 $y=1$

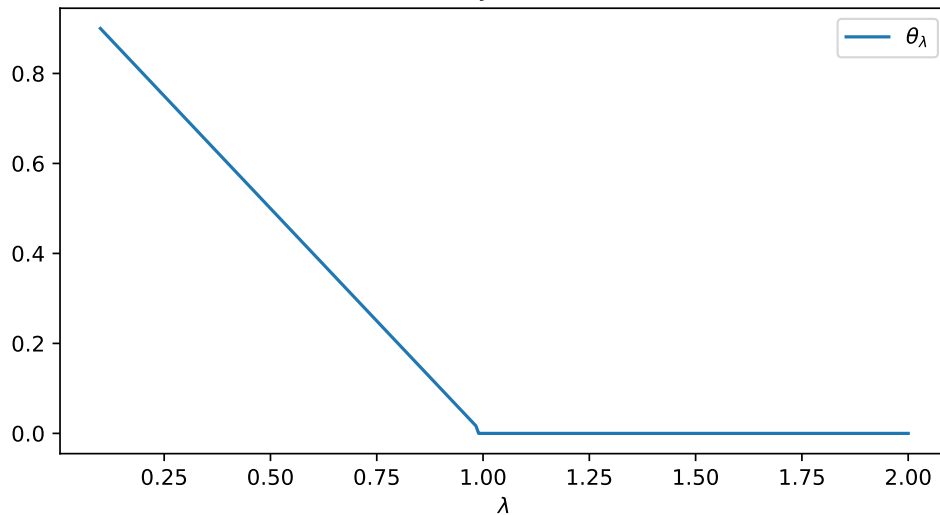


FIGURE 1 – Solution of the one-dimensional Lasso estimator

1 SOLUTION

Depending on the sign of θ , and the values of λ and y , the situation will be different. However, we need to keep in mind that we always have that $\lambda \geq 0$, as it is a regularization parameter.

1.0.1 $\theta \geq 0$

If $\theta \geq 0$, we have that

$$F_\lambda(\theta) = \frac{1}{2}(y - \theta)^2 + \lambda\theta \quad (1)$$

We can differentiate with respect to θ , and

$$F'_\lambda(\theta) = -(y - \theta) + \lambda = \theta - (y - \lambda) \quad (2)$$

In this case,

- If $\lambda \geq y$, then $y - \lambda \leq 0$, and we always have that $F'_\lambda(\theta) \geq 0$, the minimum on $\mathbb{R}_+$ is attained for $\theta = 0$, with value $\frac{1}{2}y^2$.
- If $\lambda \leq y$, then $y - \lambda \geq 0$, and we have that $F'_\lambda(\theta) \leq 0$ if $\theta \leq y - \lambda$ and $F'_\lambda(\theta) \geq 0$ if $\theta \geq y - \lambda$, the minimum on $\mathbb{R}_+$ is attained for $\theta = y - \lambda$, with value $\frac{1}{2}\lambda^2 + \lambda(y - \lambda) = \lambda(y - \frac{1}{2}\lambda)$

1.0.2 $\theta \leq 0$

If $\theta \leq 0$, we have that

$$F_\lambda(\theta) = \frac{1}{2}(y - \theta)^2 - \lambda\theta \quad (3)$$

We can differentiate with respect to θ , and

$$F'_\lambda(\theta) = -(y - \theta) - \lambda = \theta - (y + \lambda) \quad (4)$$

In this case,

- If $\lambda \geq -y$, then $y + \lambda \geq 0$, and we always have that $F'_\lambda(\theta) \leq 0$, the minimum on $\mathbb{R}_-$ is attained for $\theta = 0$, with value $\frac{1}{2}y^2$.
- If $\lambda \leq -y$, then $y + \lambda \leq 0$, and we have that $F'_\lambda(\theta) \leq 0$ if $\theta \leq y + \lambda$ and $F'_\lambda(\theta) \geq 0$ if $\theta \geq y + \lambda$, the minimum on $\mathbb{R}_-$ is attained for $\theta = y + \lambda$, with value $\frac{1}{2}\lambda^2 - \lambda(y + \lambda) = \lambda(-y - \frac{1}{2}\lambda)$

1.1 Applications

Let us consider some values of y and plot the θ_λ as a function of λ .

1.1.1 $y = 1$

In this case :

- If $\lambda \leq 1$,
 - the optimal $\theta \in \mathbb{R}_+$ is $\theta_+ = 1 - \lambda$ with value $\lambda(1 - \frac{1}{2}\lambda)$.
 - the optimal $\theta \in \mathbb{R}_-$ is $\theta_- = 0$ with value $\frac{1}{2}$.

Let us show that is $0 \leq \lambda \leq 1$, then $g(\lambda) = \lambda(1 - \frac{1}{2}\lambda) \leq \frac{1}{2}$. We have that $g(\lambda) = \lambda - \frac{1}{2}\lambda^2$, hence $g'(\lambda) = 1 - \lambda \geq 0$. The maximum value of $g(\lambda)$ is thus attained in $\lambda = 1$, and is equal to $\frac{1}{2}$.

Overall, we have

$$\theta_\lambda = 1 - \lambda \quad (5)$$

and

$$F_\lambda(\theta_\lambda) = \lambda(1 - \frac{1}{2}\lambda) \quad (6)$$

- If $\lambda \geq 1$,
 - the optimal $\theta \in \mathbb{R}_+$ is $\theta_+ = 0$ with value $\frac{1}{2}$.
 - the optimal $\theta \in \mathbb{R}_-$ is $\theta_- = 0$ with value $\frac{1}{2}$.

Overall

$$\theta_\lambda = 0 \quad (7)$$

and

$$F_\lambda(\theta_\lambda) = \frac{1}{2} \quad (8)$$